Project Report:- Deadlock Detection and Recovery Toolkit

Project Objectives

1. Understand Deadlock Concepts

To study and demonstrate the concept of deadlocks in Operating Systems, including necessary conditions like mutual exclusion, hold and wait, no preemption, and circular wait.

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2. Detect Deadlocks Efficiently

To design and implement algorithms (such as Resource Allocation Graph or Banker's Algorithm) that can identify deadlocks in a system dynamically.

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3. Simulate Resource Allocation

To create a system that simulates multiple processes competing for limited resources, helping visualize how deadlocks occur.

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4. Implement Recovery Techniques

To develop mechanisms to resolve deadlocks using methods like:

- Process termination
- Resource preemption
- Rollback techniques

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5. Provide Visualization Interface

To build a user-friendly interface (GUI/Web app) that visually represents:

- Processes
- Resources
- Allocation graphs
- Deadlock conditions

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6. Improve System Performance Awareness

To analyze how deadlocks impact system performance and demonstrate strategies to avoid system failure.

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7. Educational Tool Development

To create a learning tool that helps students understand complex OS concepts through simulation and visualization.

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8. Real-Time Monitoring

To implement real-time monitoring of system state and dynamically detect unsafe conditions.

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9. Compare Algorithms

To compare different deadlock detection and recovery approaches in terms of efficiency and resource utilization.

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10. Practical Implementation Skills

To gain hands-on experience in:

- Programming (Python/Java/Flask)
- System design
- Algorithm implementation

Conclusion

The **Deadlock Detection and Recovery Toolkit** demonstrates how deadlocks occur and how they can be detected and resolved using efficient algorithms. It simulates resource allocation among processes and applies recovery techniques like process termination and resource preemption. Overall, the project enhances understanding of operating system concepts and provides a practical approach to managing system stability.